

Acute Pancreatitis: Clinical and Morphologic Classification

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See the slide presentation [here](#).

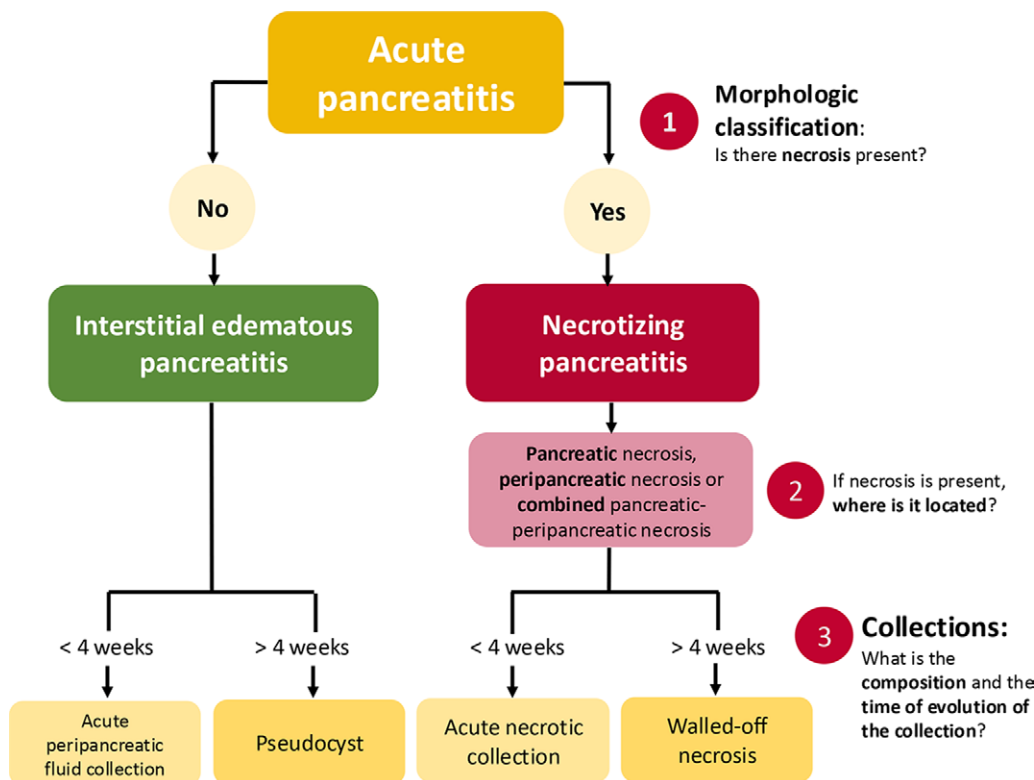
Acute pancreatitis is a dynamic and complex pathologic condition with a variable course and a wide spectrum of clinical and imaging manifestations. The accompanying slide presentation discusses the clinical and morphologic classifications of acute pancreatitis.

The 2012 revised Atlanta classification, developed by the Acute Pancreatitis Classification Working Group after a comprehensive review and modification of the original 1992 system, stands as the most widely used clinically based classification system among clinical physicians and radiologists. The goal of the revision was to integrate the latest concepts to understand the disease and mitigate ambiguity and confusion surrounding certain terms in the original classification. The revision established a clinical and radiologic nomenclature with more precise definitions and standardized terminology for describing and categorizing the manifestations associated with acute pancreatitis. The use of standardized and disease-specific language is fundamental because it al-

lows effective, precise, and objective communication among members of a multidisciplinary team.

The consensus classification defined diagnostic criteria for acute pancreatitis and subdivided the clinical course of the disease into an early phase (<1 week) and a late phase (>1 week), as well as categorized its severity into mild, moderate, and severe, based on the presence and duration of multiorgan failure and other local and systemic complications.

Given the increasing importance of imaging evaluation for the diagnosis and treatment of acute pancreatitis, the revised Atlanta classification primarily relies on morphologic imaging findings to define the various manifestations of the disease (Fig 1), predominantly through contrast-enhanced CT. This method is chosen as the primary tool for assessing the imaging-based criteria because of its high precision and widespread availability, which make it a practical option for severely ill patients. Thus, acute pancreatitis is classified into two morphologic subtypes based on the absence or presence





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TEACHING POINTS

- Developed by experts, the revised Atlanta classification is the only system accepted by clinicians and radiologists to define and categorize the manifestations of acute pancreatitis.
- The morphologic classification of acute pancreatitis and the definition of local collections depend on the presence or absence of necrosis. Thus, pancreatitis is divided into interstitial edematous pancreatitis and necrotizing pancreatitis, and collections can be acute peripancreatic fluid collections and pseudocysts, if they contain only fluid, and acute necrotic collection and walled-off necrosis, if they contain fluid and necrotic debris.
- In addition to local collections, other local complications should be identified, particularly in the context of necrotizing pancreatitis, such as infection, pancreatic-enteric fistulas, biliary obstruction, vascular complications, and mass effect.

of necrosis in the pancreatic parenchyma and/or peripancreatic tissues: interstitial edematous pancreatitis and necrotizing pancreatitis, respectively (Fig 2). Moreover, associated local collections are classified into four subtypes based on their composition and presence since the onset of pancreatitis: acute peripancreatic fluid collections and pseudocysts that occur in interstitial edematous pancreatitis and contain only fluid; and acute necrotic collections and walled-off necrosis that occur in necrotizing pancreatitis and contain varying amounts of fluid and solid components.

In addition to defining local collections, precise identification and description of other local complications (particularly in cases of necrotizing pancreatitis), such as vascular, gastrointestinal, biliary, inflammatory, or infectious complications, also are essential because they can exacerbate the patient's condition and necessitate specific therapeutic management.

Although the use of standardized definitions facilitates the categorization and description of manifestations associated with acute pancreatitis, the more complex forms of the disease may present a formidable interpretive challenge for radiologists.

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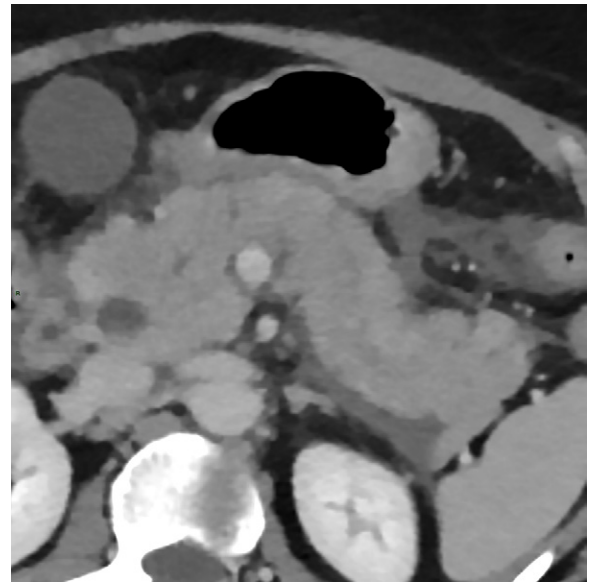


Figure 2. Axial contrast-enhanced CT image in a 36-year-old man with interstitial edematous pancreatitis. This morphologic subtype is characterized by focal or diffuse pancreatic enlargement with no unenhanced areas.

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Suggested Readings

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